

JAVA 9th Experiment - Hibernate CRUD Application

Dear Students, upload your JAVA 9th Experiment in this link by 20-04-2025.

1. Develop a Hibernate-based application to perform CRUD (Create, Read, Update, Delete) operations on a Student entity using Hibernate ORM with MySQL.

Requirements:

1. Configure Hibernate using hibernate.cfg.xml.
2. Create an Entity class (Student.java) with attributes: id, name, and age.
3. Implement Hibernate SessionFactory to perform CRUD operations.
4. Test the CRUD functionality with sample data.

Complete Code Solution:

1. hibernate.cfg.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE hibernate-configuration PUBLIC
    "-//Hibernate/Hibernate Configuration DTD 3.0//EN"
    "http://www.hibernate.org/dtd/hibernate-configuration-3.0.dtd">
<hibernate-configuration>
    <session-factory>
        <property name="connection.driver_class">com.mysql.cj.jdbc.Driver</property>
        <property name="connection.url">jdbc:mysql://localhost:3306/student_db</property>
        <property name="connection.username">root</property>
        <property name="connection.password">your_password</property>
        <property name="dialect">org.hibernate.dialect.MySQL8Dialect</property>
        <property name="show_sql">true</property>
        <property name="hbm2ddl.auto">update</property>
        <mapping class="com.example.Student"/>
    </session-factory>
</hibernate-configuration>
```

2. Student.java

```
package com.example;

import jakarta.persistence.*;
```

```

@Entity
@Table(name = "students")
public class Student {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private int id;

    private String name;
    private int age;

    public Student() {}

    public Student(String name, int age) {
        this.name = name;
        this.age = age;
    }

    public int getId() { return id; }
    public void setId(int id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public int getAge() { return age; }
    public void setAge(int age) { this.age = age; }
}

```

3. HibernateUtil.java

```

package com.example;

import org.hibernate.SessionFactory;
import org.hibernate.cfg.Configuration;

public class HibernateUtil {

    private static final SessionFactory sessionFactory;

    static {
        try {
            sessionFactory = new
Configuration().configure("hibernate.cfg.xml").buildSessionFactory();
        } catch (Throwable ex) {
            throw new ExceptionInInitializerError(ex);
        }
    }

    public static SessionFactory getSessionFactory() {
        return sessionFactory;
    }
}

```

4. StudentDAO.java

```

package com.example;

import org.hibernate.Session;
import org.hibernate.Transaction;

public class StudentDAO {

    public void createStudent(Student student) {
        Transaction tx = null;
        try (Session session = HibernateUtil.getSessionFactory().openSession()) {
            tx = session.beginTransaction();
            session.save(student);
            tx.commit();
        } catch (Exception e) {
            if (tx != null) tx.rollback();
            e.printStackTrace();
        }
    }

    public Student getStudent(int id) {
        try (Session session = HibernateUtil.getSessionFactory().openSession()) {
            return session.get(Student.class, id);
        }
    }

    public void updateStudent(Student student) {
        Transaction tx = null;
        try (Session session = HibernateUtil.getSessionFactory().openSession()) {
            tx = session.beginTransaction();
            session.update(student);
            tx.commit();
        } catch (Exception e) {
            if (tx != null) tx.rollback();
            e.printStackTrace();
        }
    }

    public void deleteStudent(int id) {
        Transaction tx = null;
        try (Session session = HibernateUtil.getSessionFactory().openSession()) {
            Student student = session.get(Student.class, id);
            if (student != null) {
                tx = session.beginTransaction();
                session.delete(student);
                tx.commit();
            }
        } catch (Exception e) {
            if (tx != null) tx.rollback();
            e.printStackTrace();
        }
    }
}

```

5. Main.java

```
package com.example;

public class Main {
    public static void main(String[] args) {
        StudentDAO dao = new StudentDAO();

        Student s1 = new Student("Ravi", 21);
        dao.createStudent(s1);

        Student fetched = dao.getStudent(s1.getId());
        System.out.println("Fetched Student: " + fetched.getName());

        fetched.setAge(22);
        dao.updateStudent(fetched);

        dao.deleteStudent(fetched.getId());
    }
}
```